

AWS Cloud Foundations for Machine Learning

How AWS infrastructure, services, responsibility, and architecture support ML workloads

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Cloud computing changes how ML resources are obtained

Traditional infrastructure

Organizations purchase servers, estimate peak capacity, maintain hardware, and wait for new resources.

AWS Cloud

Teams request managed storage, compute, databases, and AI services when needed—and release them when demand changes.

AWS infrastructure separates location, scale, and resilience

01 Regions

Geographic areas used to place workloads near users and meet regulatory needs.

02 Availability Zones

Independent locations within a Region that support fault-tolerant designs.

03 Edge locations

Distributed sites that bring content and selected services closer to users.

04 Global services

Services such as IAM operate across Regions while still controlling regional resources.

An AWS ML solution combines four service layers

01 Data

Amazon S3, databases, streaming services, Glue, and analytics platforms supply usable information.

02 Build

SageMaker AI and Amazon Bedrock support traditional ML and generative AI development.

03 Deploy

Endpoints, batch jobs, containers, and orchestration services deliver predictions.

04 Operate

CloudWatch, IAM, KMS, CloudTrail, and cost tools keep solutions observable and controlled.

AWS secures the cloud; customers secure their workloads

AWS responsibility

Physical facilities, hardware, networking, and the infrastructure that runs AWS services.

Customer responsibility

Data, identities, permissions, configurations, encryption choices, and application behavior.

Good cloud decisions balance six architectural qualities

01 Security

Protect data, identities, models, endpoints, and logs.

02 Reliability

Recover from failure and meet availability requirements.

03 Performance

Choose resources that satisfy throughput and latency needs.

04 Cost & operations

Control spending while making workloads observable and maintainable.

The cloud foundation gives us a decision framework

Where should the workload run?
Which services provide the needed capability?
Who manages each security responsibility?
How will the design balance cost and performance?

NEXT

Part 2: Machine
learning foundations